

ALIGNER BOX – FEATURES

1. **Automated aligner detection** – whether the aligner is placed inside the box or not and also whether it is placed in the oral cavity or place somewhere outside the oral cavity
2. **IR based communication and detection** – with the aligner which contain ir sensor
3. **External storage -** to store the datas of aligner worn or removed along with the time and date of upto **150 to 200 days (minimum)** and should **work offline** or without any phone near to it should work on its own
4. **RTC module –** to get the real time and date (**backup battery** to prevent the reset of the time & date in case the user didn't charged the aligner box)
5. **Main battery –** it should be **rechargeable** using the normal **C type charging port**
6. **Bluetooth / Wifi module –** to transfer the datas from the aligner to the app
7. **Buzzer –**
 - a. to indicate the user that aligner is outside the box and also not worn
 - b. to indicate that the aligner is kept inside the box for long time (like indicating the user after lunch or dinner they forgot to worn the aligner and the sound of the buzzer should differ for 30min, and 1hr, after 30min the buzzer should sound for every 30min)
 - c. to indicate whether aligner box is going to switch off without the power (**low battery**)
 - d. data transfer completed
 - e. aligner **communication failed** with the aligner box
 - f. aligner box communication failed with the app

- g. box reset
- h. charging completed

8. **LED Indicator** – to indicate the status like charging, low battery, synchronization, error, aligner detected & connected with the box (multicolour led indicator, the mentioned colours are not mandatory just for idea I have added the colours)

- a. aligner detected - **Green**
- b. data transfer from aligner to box or from box to the app when connected – **Blue**
- c. aligner out for long time (indicate along with buzzer) - **Yellow**
- d. critical issues – low battery, communication error (along with the buzzer) - **Red**
- e. charging indicator & colour should be changed after the charging completed along with buzzer for full charge - **White**
- f. Pairing mode/setup mode – **Any colour – Blinking**